

Michal Malček — AI Engineer / Explorer / SEO

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Interactive version: <https://misomalcek.github.io/cyber-cv/> / GitHub: <https://github.com/misomalcek>

In one paragraph

Twenty years across both sides of software delivery — ERP testing at nineteen, Google in Prague and Dublin at twenty-four, eight years proving things with measurement in a market that punishes wishful thinking. In the last year that turned into AI engineering: one production system delivered and handed over, one running daily on the machine under my desk. Bachelor's in Applied Informatics, hooked to cybernetics since high school, one three-day AI generalist course — everything else learned from the systems themselves.

Two systems

Team Brain — a multi-source SEO analytics platform an agency team used daily. Fifteen users, five live data sources joined behind one query surface, one 8 GB VPS. **93.9% tool success over 1,352 real calls** by non-engineers. Six specialists with the model chosen per role (Sonnet for analysis, Haiku for generation), five routing signals tried in order, nine autonomous jobs on cron. 49.4 M input tokens over five months — about **\$30/month** in Claude API spend, because snapshots answer first and the live API is the fallback. Built on the Anthropic SDK, delivered with nine handover documents.

Factorium — a local-first desktop environment where a person and two models share one memory. One Apple M4, 24 GB, **no cloud inference**. One Postgres holding pgvector, two Apache AGE graphs and a TimescaleDB hypertable, with Qdrant alongside. 23 tool modules behind one registry; the local model gets a deliberately smaller surface because every extra schema is prompt tax on a 12B. Six-layer retrieval fused by rank. **186 source files, 563 commits, build passes at 2,911 modules, 23/23 smoke tests**. Open source at release.

Research

Three papers and an unedited audit of them by the local model, all written from stored records rather than recollection.

- **Rules retrieved by their own wording: 1 in 5. Stored as the situations that trigger them: 10 in 10.**
- **Asked which of its own rules carried real information, the model misclassified 7 of 8** — including three that had changed its decision minutes earlier. Value is measured by behaviour, never by asking.
- **1,512 of 3,039 entities silently lost their centrality tier** after one ordinary session. A derived global property is a cache, not a field.
- A selection defect found after publication: the measured rule set had been seeded by a **filename pattern**, which excluded the most-used rule in the system. Excluded rules average graph degree 7.1

against 3.9 for included ones.

Experience

2024 – 2026 · AI Engineer (Team Brain, Factorium) Two systems, described above.

2023 – 2026 · SEO & Content Lead — msg life Slovakia · Swiss Re (parallel, 3 months) Swiss Re: keyword and synonym coverage for the Sigma annual report, readership ~1,500 → **4,500**. msg life: first large keyword clusters with Gemini + Ahrefs, **traffic doubled inside a year** on WordPress. Ahrefs workshops at both.

2019 – 2023 · Senior SEO Manager — Lottoland, Gibraltar 50% management (4 internal, 5 external, 3 markets), 50% hands-on. Three internal SEO trainings.

2015 – 2019 · SEO Manager — Lottostar24 · Tippland Hamburg German traffic **zero** → **30k monthly sessions**, 100k across satellites. 0.9% vested equity. Two systems in Laravel with one developer — a result pipeline recomputing statistics after every draw into live article text, and a distribution layer feeding a network without duplicates. **Ran unattended for two years and still ranks**. Server-side prerendering for Angular, years before frameworks made it routine.

Not one domain penalised and no legal claim in eight years, in a vertical where competitors kept a floor of lawyers.

2012 – now · Founder, My Marketing s.r.o. · co-founded a CEE sportswear e-commerce reaching €50k monthly turnover in year one.

2010 – 2011 · Google, Prague and Dublin.

Education

Bachelor of Applied Informatics, University of Economics Prague (avg 1.6) · CEMS International Management, taught in English (avg 1.7) · Information Systems & Technologies, 41 ECTS · Erasmus at Corvinus Budapest, A in every subject · Top 10 student of ~500.

Every document is in the repository at <https://misomalcek.github.io/cyber-cv/credentials/> — no sign-in, no dead links.

Languages

Slovak and Czech native · English advanced (a full master's studied in it) · Russian A2, third worldwide in a St Petersburg University olympiad · German A1, Goethe-Institut.